Graphic design

Simplicity

- "Perfection is achieved not when there is nothing more to add, but when there is nothing left to take away."
  - Antoine de St-Exupery
- "Simplicity does not mean the absence of any decor… It only means that the decor should belong intimately to the design proper, and that anything foreign to it should be taken away."
  - Paul Jacques Grillo
- "Keep it simple, stupid." (KISS)
- "Less is more."
- "When in doubt, leave it out."

- Simplicity forces you to have a good reason for everything you add

Techniques for Simplicity: Reduction

- Remove inessential elements

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A photograph of a pair of scissors can't possibly work as a 32x32 pixel icon

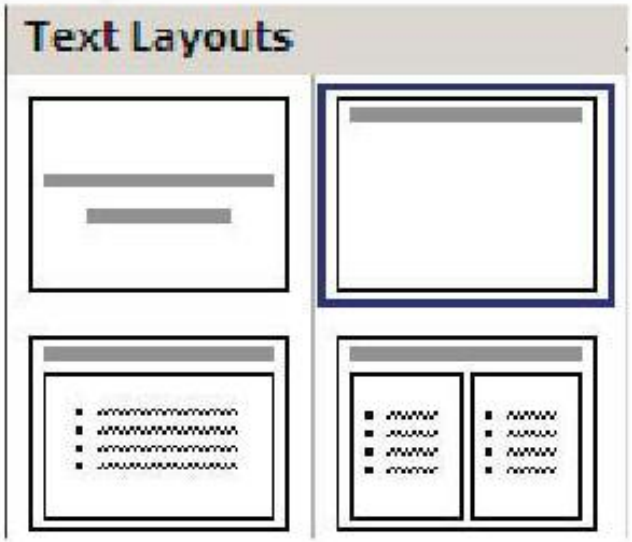
This image is in the public domain. Source: US Department of Transportation.

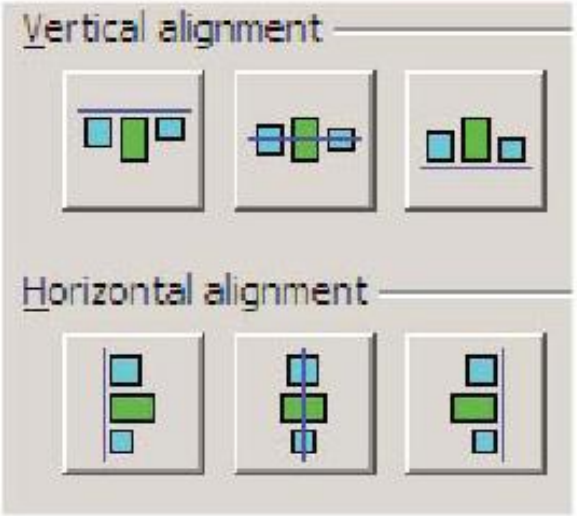
No element remains that can be removed from it without destroying its meaning

Techniques for Simplicity: Regularity

- Use a regular pattern
- Limit inessential variation among elements

PowerPoint's Text Layouts menu shows both reduction (minimalist icons representing each layout) and regularity
Titles and bullet lists are shown the same way





Irregularities in design will be magnified in the user's eyes and assigned meaning and significance

Techniques for Simplicity: Double-Duty

- Combine elements for leverage
  - Find a way for one element to play multiple roles

title bar

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scrollbar thumb

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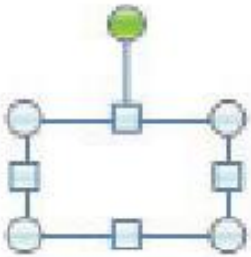
selection handles



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help prompt

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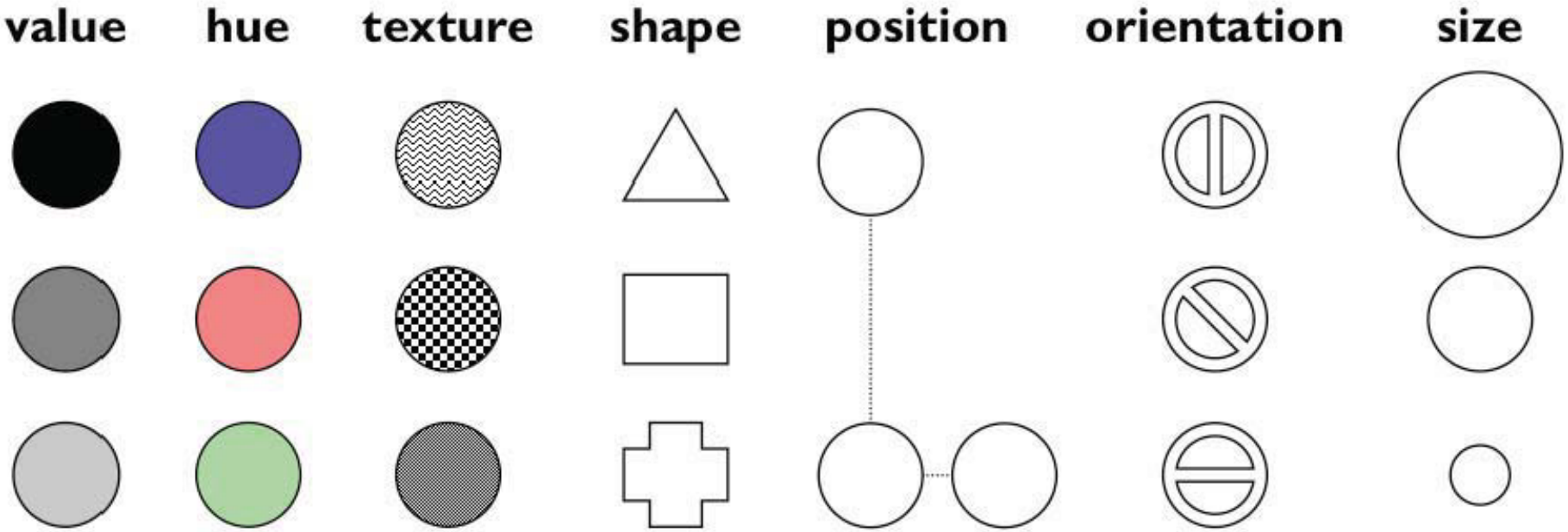
Contrast & Visual Variables

- Contrast encodes information along visual dimensions



Contrast is the irregularity in a design that communicates information or makes elements stand out. Simplicity says we should eliminate **unimportant differences. Once we've decided that a difference** is important, we should choose the dimension and degree of contrast in such a way that the difference is salient, easily perceptible, and appropriate

Characteristics of Visual Variables

- Scale = kinds of comparisons possible
  - Nominal (=)
    - All variables
  - Ordered (<, >)
    - Ordered: position, size, value, texture granularity
    - Not ordered: orientation, hue, shape
  - Quantitative (amount of difference)
    - Quantitative: position, size
    - Not quantitative: value, texture, orientation, hue, shape
- Length = number of distinguishable levels
  - Shape is very long (infinite variety)
  - Position is long and fine-grained
  - Orientation is very short (~ 4 levels)
  - Other variables are in between (~ 10 levels)

Techniques for Contrast

- Choose appropriate visual variables
- Use as much length as possible
- Sharpen distinctions for easier perception
  - Multiplicative scaling, not additive
  - Redundant coding where needed
  - Cartoonish exaggeration where needed
- Use the "squint test"

full width of the window; smallest and the largest sizes

title not only larger (size), but centered (position), bold (value), and distinct color

Close one eye and squint the other, to disrupt your focus. Whatever distinctions you can still make out will be visible "at a glance."

Choosing Visual Variables for a Display

Subject	Sender	Date
Содействие в трудоустройстве.	chao	10/15/2004 4:26...
Автовладельцам	АвтоГранд	10/15/2004 4:45...
Обучение теннису	eliot	10/15/2004 7:16 AM
PITTSBURGH PA Silverton Home Services for...	Erica Gallenbeck	10/15/2004 7:21...
156 - 00 - 00 наш &#1...	XjXFXLXmXgX@tdb.com	10/15/2004 10:4...
156-00-00	hucksterEOFIN	10/15/2004 11:12 ...
A Library A Dream...	Arthur GuoBin Yin	10/15/2004 6:38...
SAVE 20% on holiday cards by shopping early	Snapfish	5:18 AM
How are you	Анисимов К.И.	11:24 AM

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- Spam flag: nominal, 2 levels (spam or not)
- Subject: nominal (but can be ordered alphabetically), infinite (but maybe only ~100 are active)
- Sender: nominal (but can be ordered alphabetically), infinite (but maybe ~100 people you know + everybody else are useful simplifications)
- Unread flag: nominal, 2 levels (read or unread)
- Date: quantitative (but maybe ordered is all that matters), infinite (but maybe only ~10 levels matter: today, this week, this month, this year, older)

Contrast in Publication Styles

Title

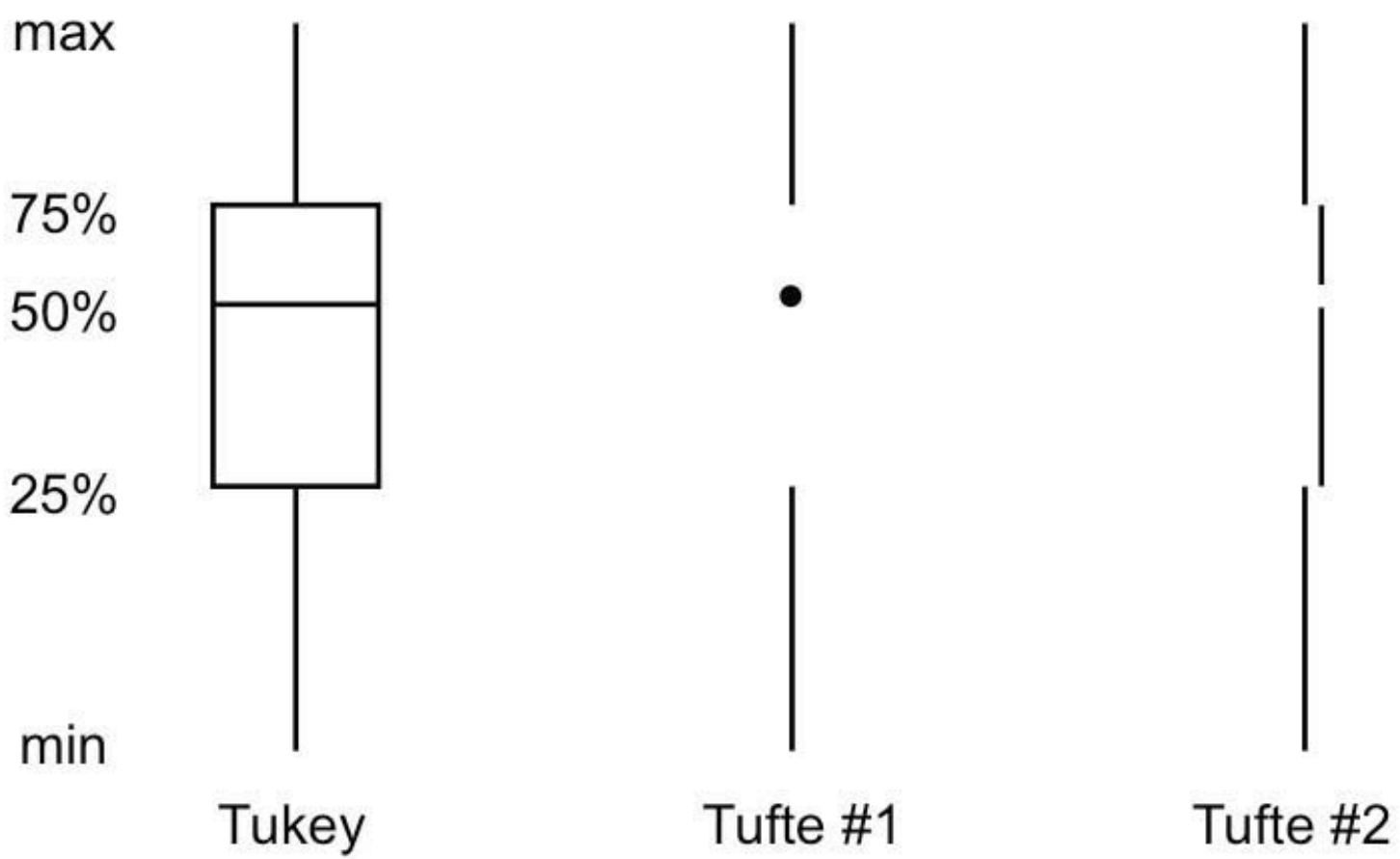
Heading

This is body text. It's smaller than the heading, lighter in weight, and longer in line length. We've also changed its shape to a serif font, because serifs make small text easier to read. Redundant encoding produces an effective contrast that makes it easy to scan the headings and distinguish headings from body text.[1]

Figure 1. This is a caption, which is smaller than body text, and set off by position, centering, and line length.

[1]This is a footnote. It's even smaller, and positioned at the bottom of the page.

Simplicity vs. Contrast

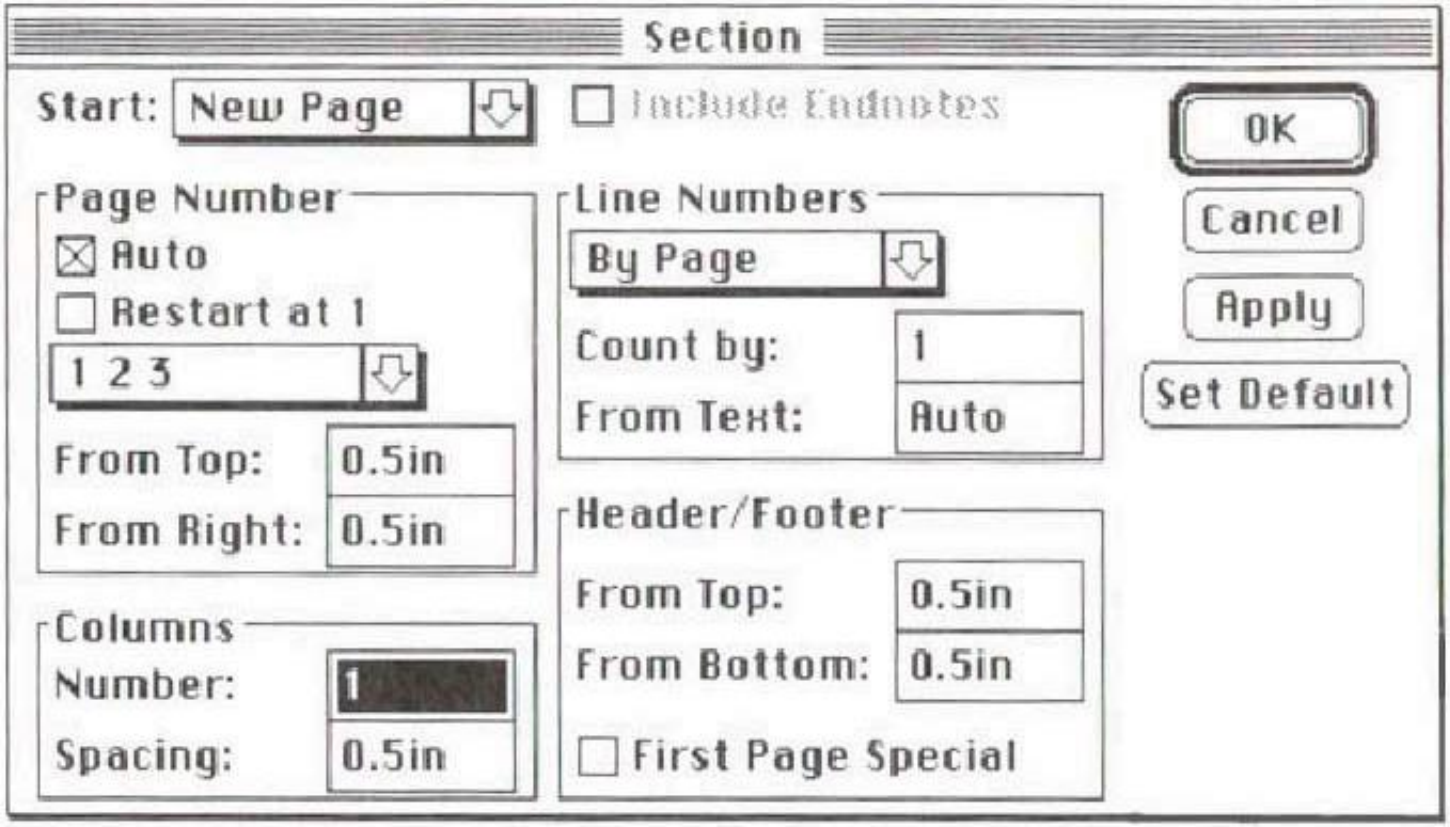


simplicity is taken too far, and contrast suffers

White Space

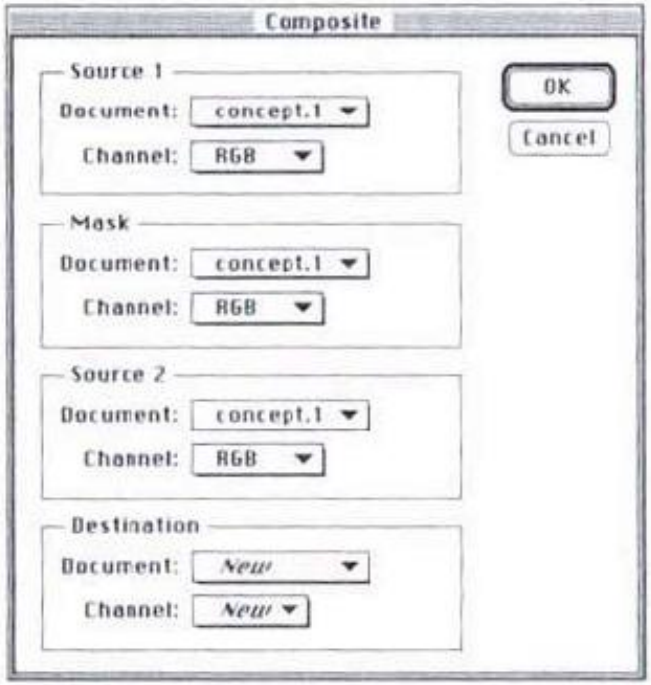
- Use white space for grouping, instead of lines
- Use margins to draw eye around design
- Integrate figure and ground
  - Object should be scaled proportionally to its background
- Don’t crowd controls together
  - Crowding creates spatial tension and inhibits scanning

Crowded Dialog

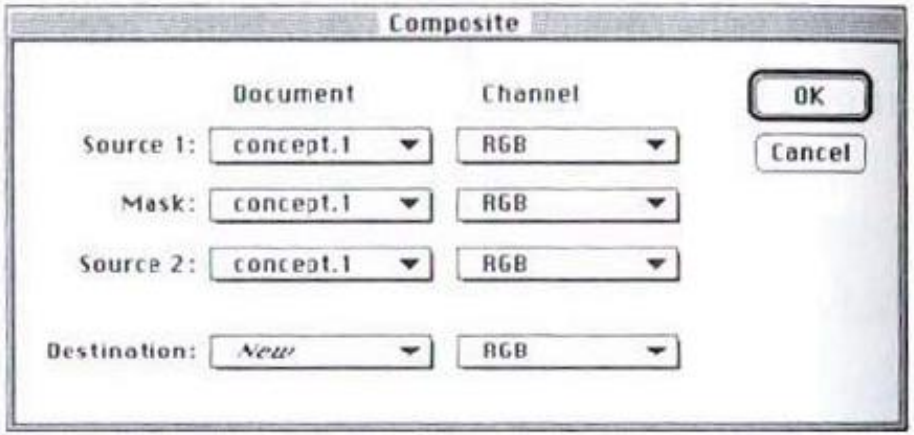


Source: Mullet & Sano, p. 110

Using White Space to Set Off Labels



(a)



(b)

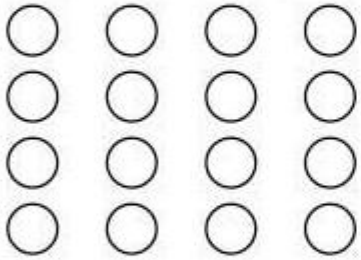
Source: Mullet & Sano, p. 96

The Gestalt Principles of Grouping

- Gestalt principles explain how eye creates a whole (*gestalt*) from parts

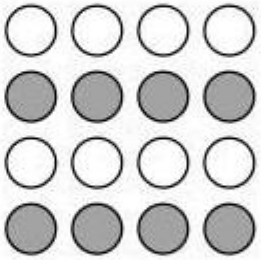
four vertical columns of circles, closer vertically than they are horizontally

proximity



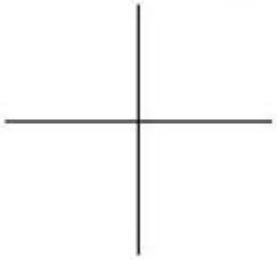
Elements with similar attributes are more likely to be grouped

similarity

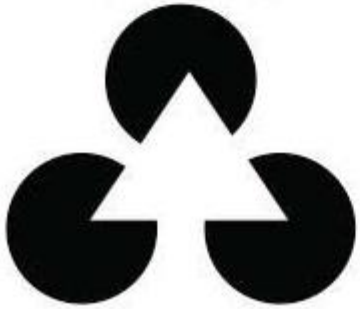


two crossing lines, rather than as four lines meeting at a point, or two right angles sharing a vertex.

continuity

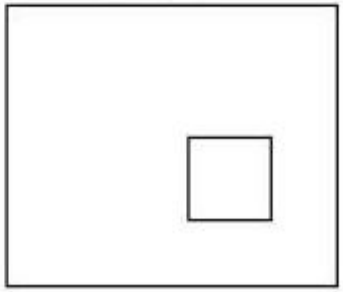


closure



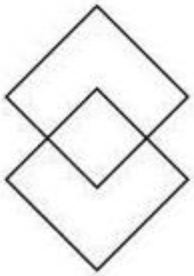
eye tends to perceive complete, closed figures

area



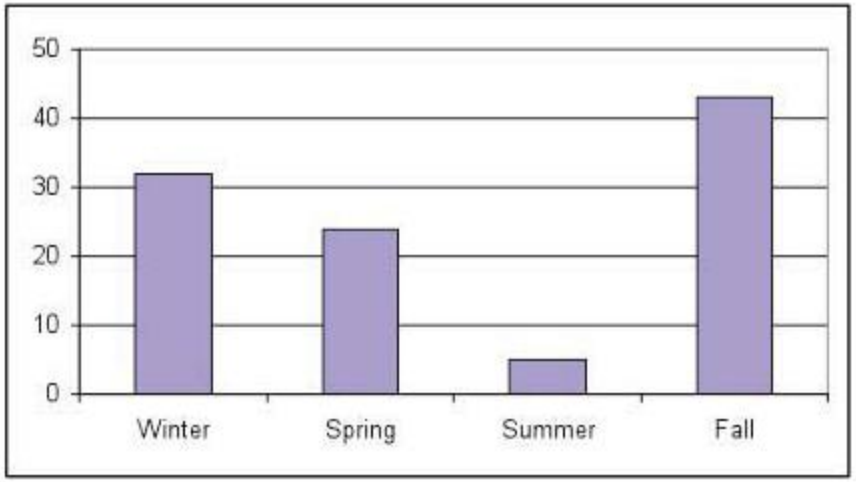
small square in front of a large square, rather than a large square with a hole cut in it

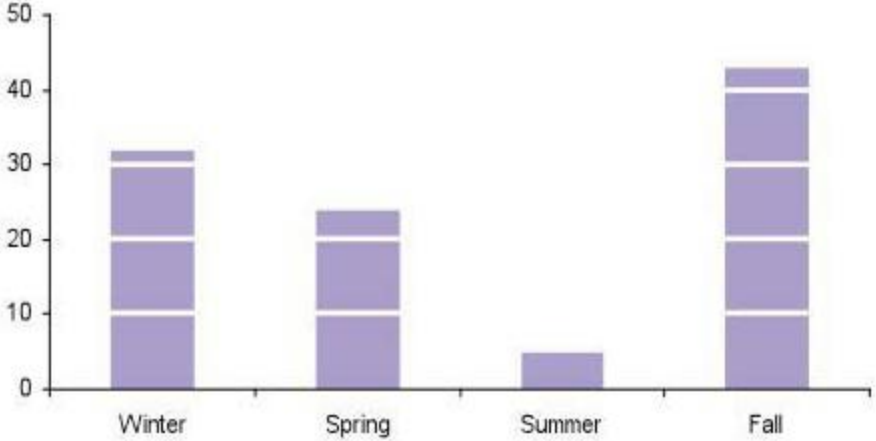
symmetry



two overlapping squares, rather than three separate polygons

White Space Avoids Visual Noise





Balance & Symmetry

- Choose an axis (usually vertical)
- Distribute elements equally around the axis
  - Equalize both mass and extent

breaks this symmetry: the stack of links for Advanced Search, Preferences, and Language Tools on the right. This slight irregularity (a kind of **contrast)** **actually helps emphasize these links slightly**

Summary

- Use contrast to emphasize important distinctions
  - Choose appropriate visual variables
  - Squint test
- Simplify unimportant distinctions
- Remember Gestalt grouping
  - Alignment, balance, symmetry